AIS – AURELIA INDUSTRIAL SYSTEMS

AIS-500 USER MANUAL

Aurelia Industrial Systems (AIS)

AIS-500 User Manual
High Performance Production System
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1. Introduction

The AIS-500 is a high-performance automated production system designed for high-throughput industrial manufacturing environments.

The system can integrate multiple robotic modules, industrial PLC control, advanced diagnostic functions, and Manufacturing Execution System (MES) connectivity.

The standard AIS-500 configuration provides a nominal maximum production capacity of:

180 units per minute.

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2. Intended Use

The AIS-500 is designed for continuous industrial production applications requiring high throughput and advanced automation.

Typical applications include:

- High-volume assembly.
- Automated production lines.
- Multi-stage robotic processes.

- Integrated production cells.
- MES-connected manufacturing environments.

The machine must only be operated within its approved configuration and documented operating limits.

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3. Safety

The AIS-500 uses a redundant safety architecture and monitored safety zones.

Operators must:

- Be trained and authorized.
- Follow the machine safety procedures.
- Verify safety-zone status.
- Verify emergency stops.
- Keep unauthorized personnel away from production areas.

Never bypass safety systems.

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4. Safety Zones

The machine may contain multiple monitored production and robotic zones.

Before automatic operation:

1. Confirm that all personnel are outside restricted zones.
2. Verify that guards are closed.
3. Confirm safety-zone status.
4. Verify emergency-stop status.
5. Confirm the safety controller reports a ready condition.

A safety-zone fault must be investigated before production resumes.

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5. Installation

Installation must be performed by AIS or an authorized installation partner.

The installation process includes verification of:

- Machine positioning.
- Electrical supply.
- Protective earth.
- Safety systems.
- Robot configuration.
- Industrial network.
- MES connectivity where applicable.
- Environmental conditions.

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6. Electrical Requirements

The standard AIS-500 electrical supply is:

400 VAC, three-phase, 50/60 Hz.

Electrical installation must be performed by qualified personnel.

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7. Startup

Before startup:

1. Inspect the machine area.
2. Confirm all safety zones are clear.
3. Verify guards.
4. Verify emergency-stop status.
5. Power on the system.
6. Wait for controller initialization.
7. Check active alarms.
8. Verify robotic-module status.
9. Verify communication status.

10. Complete the homing sequence.
11. Verify MES connectivity where required.
12. Enable automatic production.

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8. Normal Operation

The nominal maximum capacity is:

180 units/minute.

Actual production performance may be lower depending on:

- Product configuration.
- Material flow.
- Robot cycle time.
- Downstream equipment.
- MES integration.
- Production recipe.
- Tooling.

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9. MES Integration

The AIS-500 supports integration with Manufacturing Execution Systems.

MES integration may be used for:

- Production reporting.
- Production orders.
- Traceability.
- Production status.
- Operational data exchange.

Network and MES configuration must follow the customer's approved security and integration procedures.

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10. Communication

The standard AIS-500 configuration supports:

- Industrial Ethernet.
- OPC UA.
- MES integration.

Network connectivity must be configured by authorized personnel.

Unauthorized network configuration changes are not permitted.

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11. HMI and Diagnostics

The HMI provides:

- Production status.
- Machine status.
- Alarm history.
- Diagnostic information.
- Production counters.
- Maintenance information.
- System status.

Advanced diagnostic functions may be restricted to authorized service personnel.

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12. Shutdown

For normal shutdown:

1. Stop production.
2. Allow the machine to reach a safe state.
3. Disable automatic production.
4. Confirm robotic modules are stopped.
5. Review active alarms.

6. Perform the required end-of-shift inspection.
7. Follow the approved shutdown procedure.

Maintenance requires the applicable energy-isolation procedure.

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13. Preventive Maintenance

The standard AIS-500 preventive-maintenance interval is:

Every 2,500 operating hours.

Maintenance may include:

- Robot inspection.
- Mechanical inspection.
- Safety-system verification.
- Sensor inspection.
- Electrical inspection.
- PLC diagnostics.
- Network diagnostics.
- Lubrication.
- Alarm-history review.

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14. High-Throughput Operation

Because the AIS-500 is designed for high production rates, abnormal conditions should be investigated promptly.

Repeated alarms should not be ignored simply because production remains possible.

Examples of conditions requiring investigation include:

- Increasing cycle time.
- Repeated robot faults.
- Communication instability.
- Repeated safety warnings.

- Increasing product defects.
- Abnormal mechanical vibration.

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15. Troubleshooting

When an error occurs:

1. Record the error code.
2. Record the machine state.
3. Record the production impact.
4. Review the HMI alarm history.
5. Follow the approved troubleshooting procedure.
6. Escalate to Technical Support if the issue remains unresolved.

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16. Technical Specifications

Parameter	Standard Value
Product	AIS-500
Machine type	High Performance Production System
Nominal capacity	180 units/min
Power supply	400 VAC, three-phase, 50/60 Hz
Operating temperature	5–35 °C
PLC	AIS Control PLC-500
Communication	Industrial Ethernet, OPC UA, MES integration
Safety	Redundant safety architecture
Preventive maintenance	Every 2,500 operating hours

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17. Maintenance Safety

Before performing maintenance:

- Stop production.
- Isolate hazardous energy.
- Verify the machine is in a safe state.
- Follow the applicable maintenance procedure.
- Allow only qualified personnel to perform technical interventions.

Robot and electrical-system maintenance requires appropriate authorization.

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18. Technical Support

For technical support, provide:

- Machine serial number.
- Product model.
- Error code.
- Operating hours.
- Machine state.
- Production impact.
- Recent maintenance activities.
- Relevant diagnostic information.

For production-stopping incidents, the applicable critical-support process must be followed.

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19. Document Control

Version	Effective Date	Description
2.0	2025-01-10	Updated operating and safety procedures
2.1	2026-01-15	Updated diagnostics and MES documentation

Current approved version: 2.1